9b

2. **Analog Filter Design**

The transfer function *H*(s) of a low pass Butterworth Filter can be expressed in terms of the product:

$$H(s)H(-s) = \frac{1}{1+\left(\frac{s}{j\omega_c}\right)^{2N}}$$

where *N* is the order of the filter and ω_c is the filter cutoff frequency.

The poles of *H*(s)*H*(-s) are found from

$$s_p = (-1)^{\frac{1}{2N}} \cdot j\omega_c = \omega_c e^{j\frac{(2k+1+N)\pi}{2N}}, \quad k \text{ is an integer}$$

The 2N poles occur in pairs on a circle of radius ω_c in the s-plane with angular spacing of $\frac{\pi}{N}$ radians. The pole pairs are symmetrical about the $j\omega$ axis. To construct a stable causal filter, we choose the *N* poles along the left half semicircle in the s-plane for *H*(s). The remaining *N* poles in the mirrored right half plane are poles from *H*(-s).

a) Derive the transfer function H(s) of a 4$^{\text{th}}$ order Butterworth Filter. You may express your answer as the product of two second-order systems in canonical form. Let the passband edge frequency $f_p = 10\text{Hz}$. The passband edge frequency f_p (in Hz) is defined as the frequency when $|H(j\omega)|^2$ or $|H(j2\pi f)|^2$ drops to $\frac{1}{2}$ of its value at zero frequency.

Plot the Bode plots of the corresponding FRF $H(j\omega)$ of the 4$^{\text{th}}$ order Butterworth filter.

- Submit both the derivation and plot.

We can use the equation for $|H(s)|$, with
$\omega_p = 2\pi f = 20\pi$.

$$|H(s)_4| = \frac{1}{1+\left(\frac{s}{j\omega_c}\right)^8}$$

With poles spaced evenly at radius ω_c at $\frac{\pi}{4}$ on the left half of the plane. Following the equation, we have poles at

$$\frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}$$

Which is equivalent to

$$-\frac{5\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, -\frac{7\pi}{8}$$

We're only looking at the left half plane, which has poles at

$$H(s) = \frac{\omega_c^4}{(s-\omega_c e^{j\frac{5\pi}{8}})(s-\omega_c e^{j\frac{7\pi}{8}})(s-\omega_c e^{j\frac{-7\pi}{8}})(s-e^{j\frac{-5\pi}{8}})}$$

Which simplifies down to

$$H(s) = \frac{\omega_c^2}{(s-\omega_c e^{j\frac{5\pi}{8}})(s-\omega_c e^{j\frac{-5\pi}{8}})} \frac{\omega_c^2}{(s-\omega_c e^{j\frac{7\pi}{8}})(s-\omega_c e^{j\frac{-7\pi}{8}})}$$

We can simplify each of these by expanding to get

$$\boxed{H(s) = \left(\frac{\omega_c^2}{s^2 - 2\omega_c s(e^{\frac{j5\pi}{8}} + e^{\frac{-j5\pi}{8}}) + \omega_c^2}\right)\left(\frac{\omega_c^2}{s^2 - 2\omega_c s(e^{\frac{j7\pi}{8}} + e^{\frac{-j7\pi}{8}}) + \omega_c^2}\right)}$$

We can use the given equation for the magnitude
$|H(s)| = H(s)H(-s)$ to set find ω_p

$$\frac{|H(j\omega_p)|}{H(0)} = \frac{1}{\sqrt{2}}$$

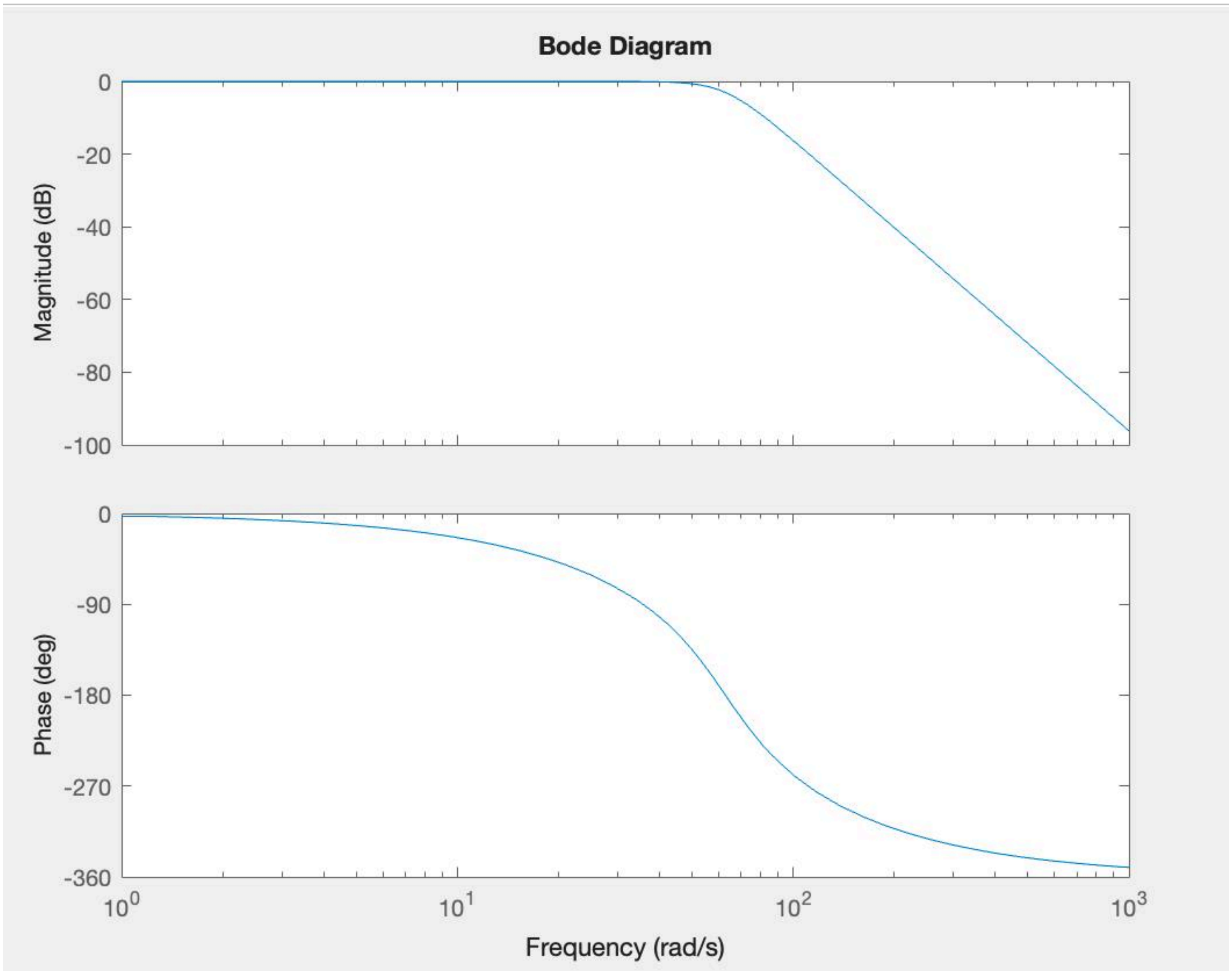
We can make this far simpler by considering

$$H(j\omega_p)H(-j\omega_p) = \frac{1}{2}$$

$$\frac{1}{1+\left(\frac{j\omega_p}{j\omega_c}\right)^8} = \frac{1}{2}$$

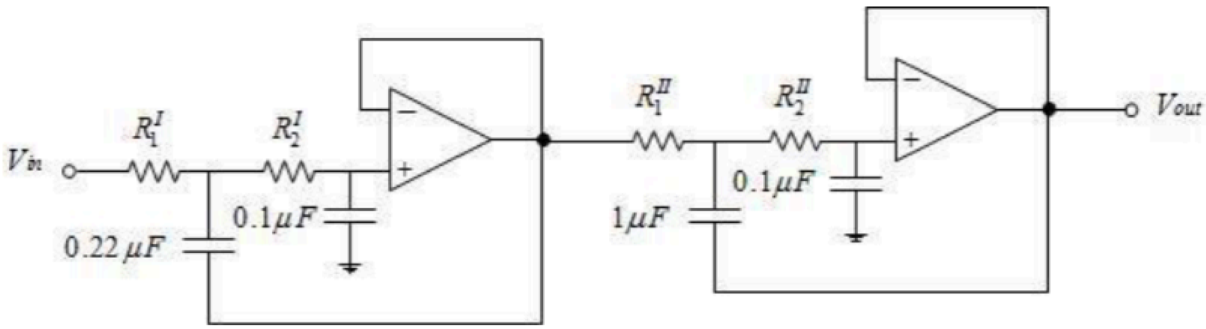
This is true when the numerator is $1 + \left(\frac{\omega_p}{\omega_c}\right)^8 = 2$

so $\boxed{\omega_c = \omega_p = 20\pi}$.



Note that the corner is at $20 * \pi$, 62.8.

b) The 4th order Butterworth Filter above is implemented with a two stage Sallen-Key circuit:



In each stage, Sallen-Key circuit transfer function $H(s)$ is given by the following equation:

$$H(s) = \frac{1}{s^2 R_1 R_2 C_1 C_2 + s(R_1+R_2)C_2 + 1}$$

Given the capacitor values shown in the diagram , i.e. we have chosen ${C_1}^I = 0.22\mu F$, ${C_2}^I = 0.1\mu F$. ${C_1}^{II} = 1\ \mu F$, ${C_2}^{II} = 0.1\mu F$.

Select "Standard" resistors ${R_1}^I, {R_2}^I, {R_1}^{II}, {R_2}^{II}$ to design the filter for $f_p \approx 10$Hz. Standard resistors come in 1.0, 1.2, 1.5, 1.8, 2.2, 2.7, 3.3, 3.9, 4.7, 5.6, 6.8, 8.2; multiplied by decade values (+/- 10%). **The solution should result in a set of more damped (higher ξ) poles in the first stage, and a less damped (lower ξ) poles in the second stage.**

- Submit the calculation of the design values ${R_1}^I, {R_2}^I, {R_1}^{II}, {R_2}^{II}$. Put the final design values in a table (calculated vs. closed standard resistors/capacitors).

We found before two second order functions. We will make one for each circuit.

$$\boxed{H(s) = \left(\frac{\omega_c^2}{s^2 - 2\omega_c s(e^{\frac{j5\pi}{8}} + e^{\frac{-j5\pi}{8}}) + \omega_c^2}\right)\left(\frac{\omega_c^2}{s^2 - 2\omega_c s(e^{\frac{j7\pi}{8}} + e^{\frac{-j7\pi}{8}}) + \omega_c^2}\right)}$$

We can rewrite this into a nice form for each, simplifying $e^{j\phi} + e^{-j\phi}$ to $2\cos(\phi)$.

$$H(s)_{\text{ part}} = \frac{1}{\frac{s^2}{\omega_n^2} - \frac{4\cos\phi}{\omega_n} + 1}$$

Pattern matching, we have

$$R_1 R_2 C_1 C_2 = \left(\frac{1}{\omega_n}\right)^2$$

and

$$(R_1 + R_2)C_2 = 4\frac{\cos\phi}{\omega_n}$$

So we have

$$
\begin{aligned}
R_1 R_2 &= \frac{1}{C_1 C_2} \frac{1}{\omega_n^2} \\
R_1 + R_2 &= \frac{4}{C_2} \frac{1}{\omega_n} \cos\phi
\end{aligned}
$$

We want to turn each into the form

$$
H(s) = \frac{1}{s^2 R_1 R_2 C_1 C_2 + s(R_1 + R_2) C_2 + 1}
$$

$$
\begin{aligned}
&\text{we have the generic form} \\
&\frac{\omega_n^2}{s^2 + 2\zeta\omega_n + \omega_n^2} \\
&= \frac{1}{\frac{1}{\omega_n^2} s^2 + \frac{2\zeta}{\omega_n} s + 1}
\end{aligned}
$$

Lets make this more concrete by tying to our system $H(s)$.

Where $\phi = \frac{5\pi}{8}$ or $\frac{7\pi}{8}$.

This tells us that

$$
\begin{aligned}
\frac{1}{\omega_n^2} &= R_1 R_2 C_1 C_2 \\
&\text{and that} \\
(R_1 + R_2) C_2 &= \frac{2\zeta}{\omega_n}
\end{aligned}
$$

We will have a system of equations with constants that has solutions with

$$
\begin{aligned}
R_1 + R_2 &= C \\
R_1 R_2 &= D \\
R_2 &= C - R_1 \\
R_1(C - R_1) &= D \\
R_1 C - R_1^2 &= D \\
R_1^2 - R_1 C + D &= 0 \\
R_1 &= \frac{C \pm \sqrt{C^2 - 4D}}{2} \\
R_2 &= \frac{D}{R_1}
\end{aligned}
$$

w_c	6.283E+01	- Cos				

phi 1	-1.963E+00	3.827E-01	H_1(j w_c)	2.761E-01		
phi 2	-2.749E+00	9.239E-01				
c11	2.200E-07		H_2(j w_c)	1.790E-01		
c12	1.000E-07		H_tot(j w_c)	4.942E-02		
c21	1.000E-06					
c22	1.000E-07					
r11	1.800E+05					
r12	6.800E+04					
r21	5.600E+05					
r22	4.700E+03					
Set 1		Actual	Goal	10% tolerance	Power of Ten	Diff
zeta	(R1+R2)C2	2.480E-02	2.436E-02	2.436E-03	-1.613E+00	-4.376E-04
w_n	sqrt(1/R1R2C1C2)	6.094E+01	6.283E+01	6.283E+00	1.798E+00	1.892E+00
C	R1+R2	2.480E+05	2.436E+05	2.436E+04	5.387E+00	-4.376E+03
D	R1R2	1.224E+10	1.151E+10	1.151E+09	1.006E+01	-7.262E+08
	R1	1.800E+05	1.795E+05	1.795E+04	5.254E+00	-5.307E+02
	R2	6.800E+04	6.415E+04	6.415E+03	4.807E+00	-3.845E+03
Set 2		Actual	Goal	10% tolerance	Power of Ten	
zeta	(R1+R2)C2	5.647E-02	5.882E-02	5.882E-03	-1.231E+00	2.346E-03
w_n	sqrt(1/R1R2C1C2)	6.164E+01	6.283E+01	6.283E+00	1.798E+00	1.193E+00
C	R1+R2	5.647E+05	5.882E+05	5.882E+04	5.769E+00	2.346E+04
D	R1R2	2.632E+09	2.533E+09	2.533E+08	9.404E+00	-9.897E+07
	R1	5.600E+05	5.838E+05	5.838E+04	5.766E+00	2.382E+04
	R2	4.700E+03	4.339E+03	4.339E+02	3.637E+00	-3.613E+02

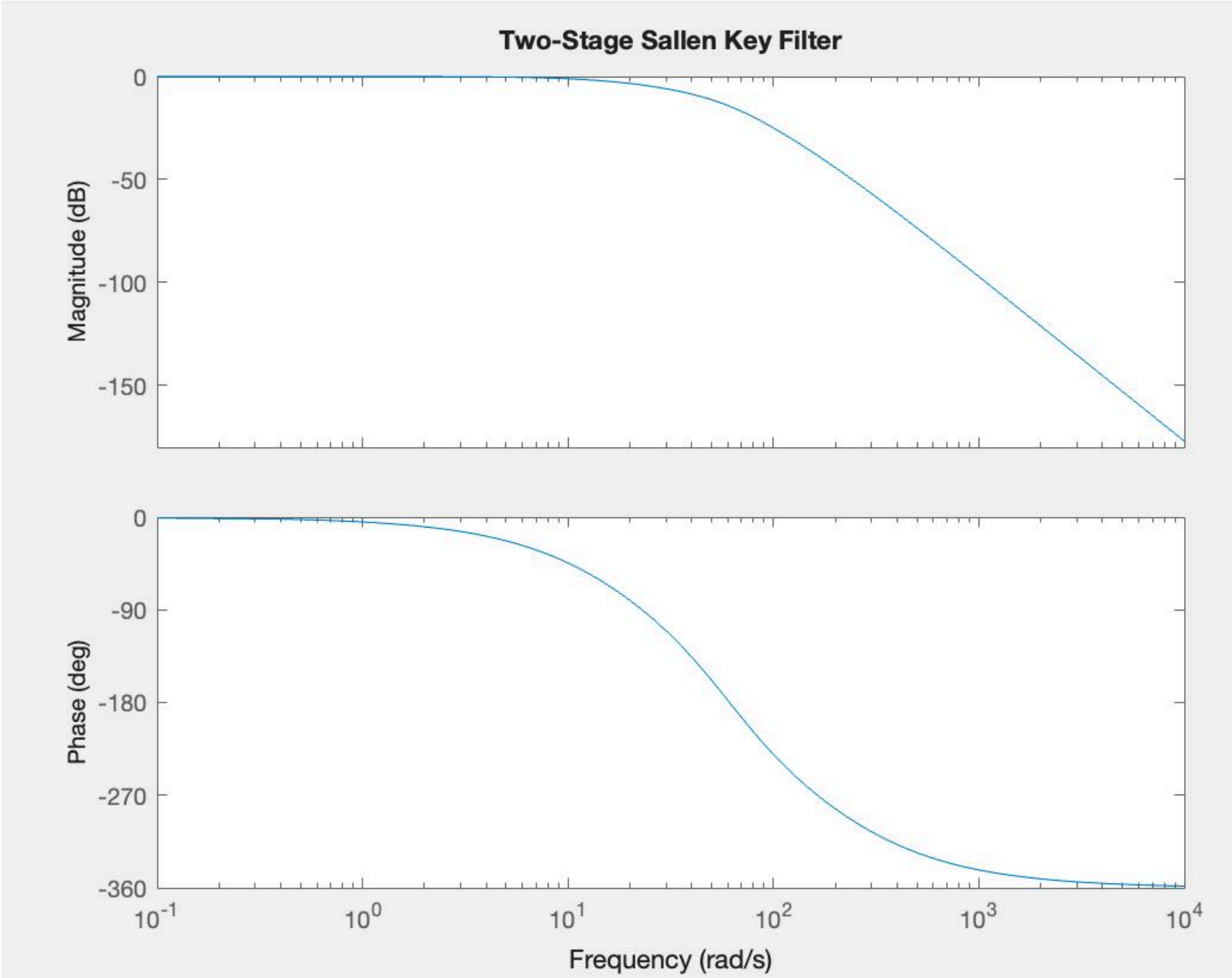
With the components for the first and second circuit given by

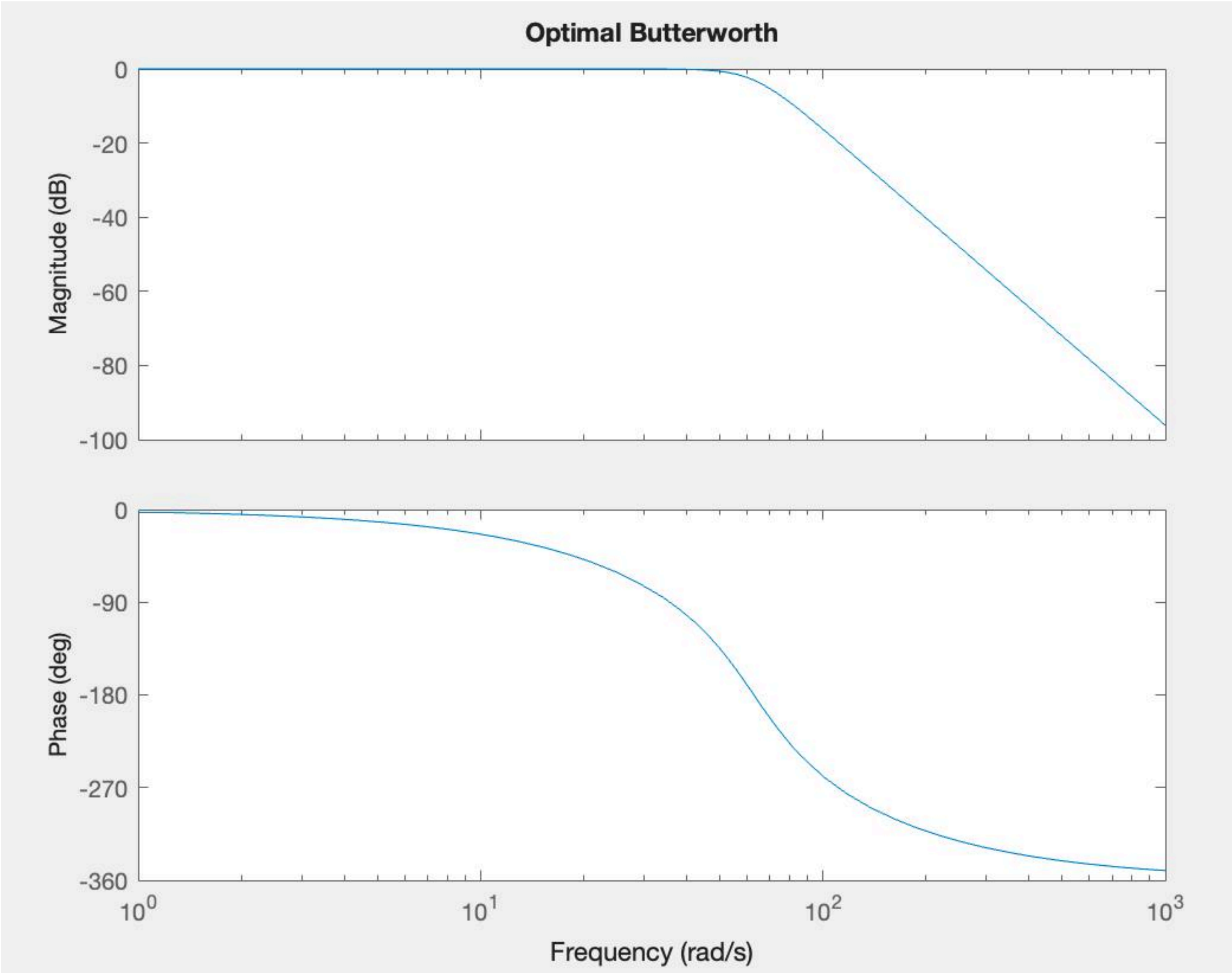
c11	**2.200E-07**	**F**
c12	1.000E-07	F
c21	1.000E-06	F
c22	1.000E-07	F
r11	1.800E+05	Ω
r12	6.800E+04	Ω
r21	5.600E+05	Ω
r22	4.700E+03	Ω

c) Create a Bode plot of the overall circuit frequency response function $|H(j\omega)|$ with the selected standard component values. Determine the passband edge frequency f_p (in Hz) from the plot.

- Submit the labeled plot

Note: Your final experimental kit will have slightly different R and C values, so you should code it up for easy adjustment and evaluate the discrepancy involved in the final implementation.


Two-Stage Sallen Key Filter
Magnitude (dB)
0
-50
-100
-150
Phase (deg)
0
-90
-180
-270
-360
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
Frequency (rad/s)



3. Digital Filter Design

A comb filter is a causal digital filter with zeros evenly spaced around the unit circle in the z-plane.

$$H_{comb}(z) = b_0(1 - z^{-M})$$

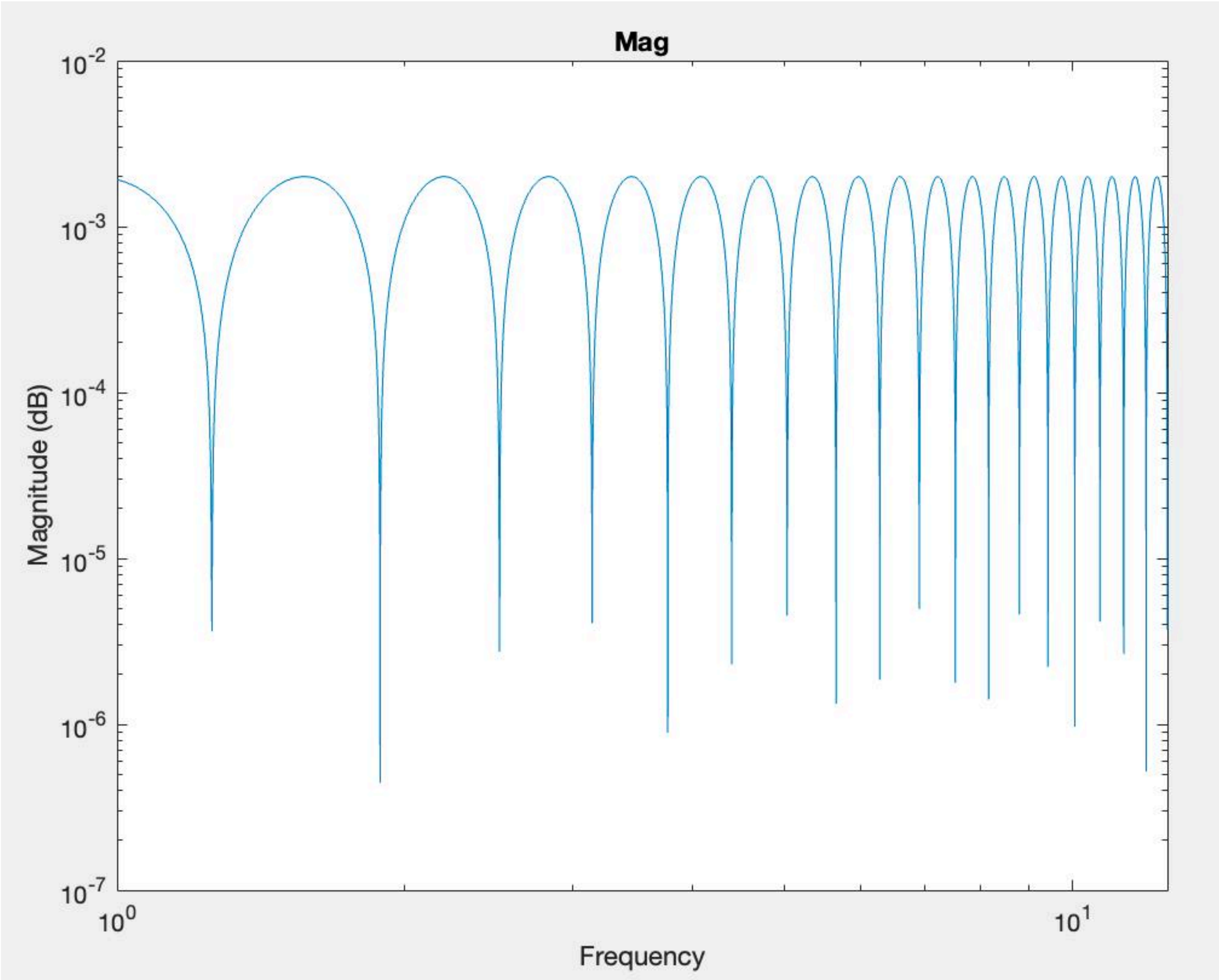
where M is the number of zeros. By selectively removing conjugate pairs of zeros, the designer can provide passbands at the selected frequencies with stopbands at the frequencies of the remaining zeros.

a) Plot the magnitude of the frequency response function $\left|H_{comb}(e^{j\hat{\omega}})\right|$ of an M=10 comb filter

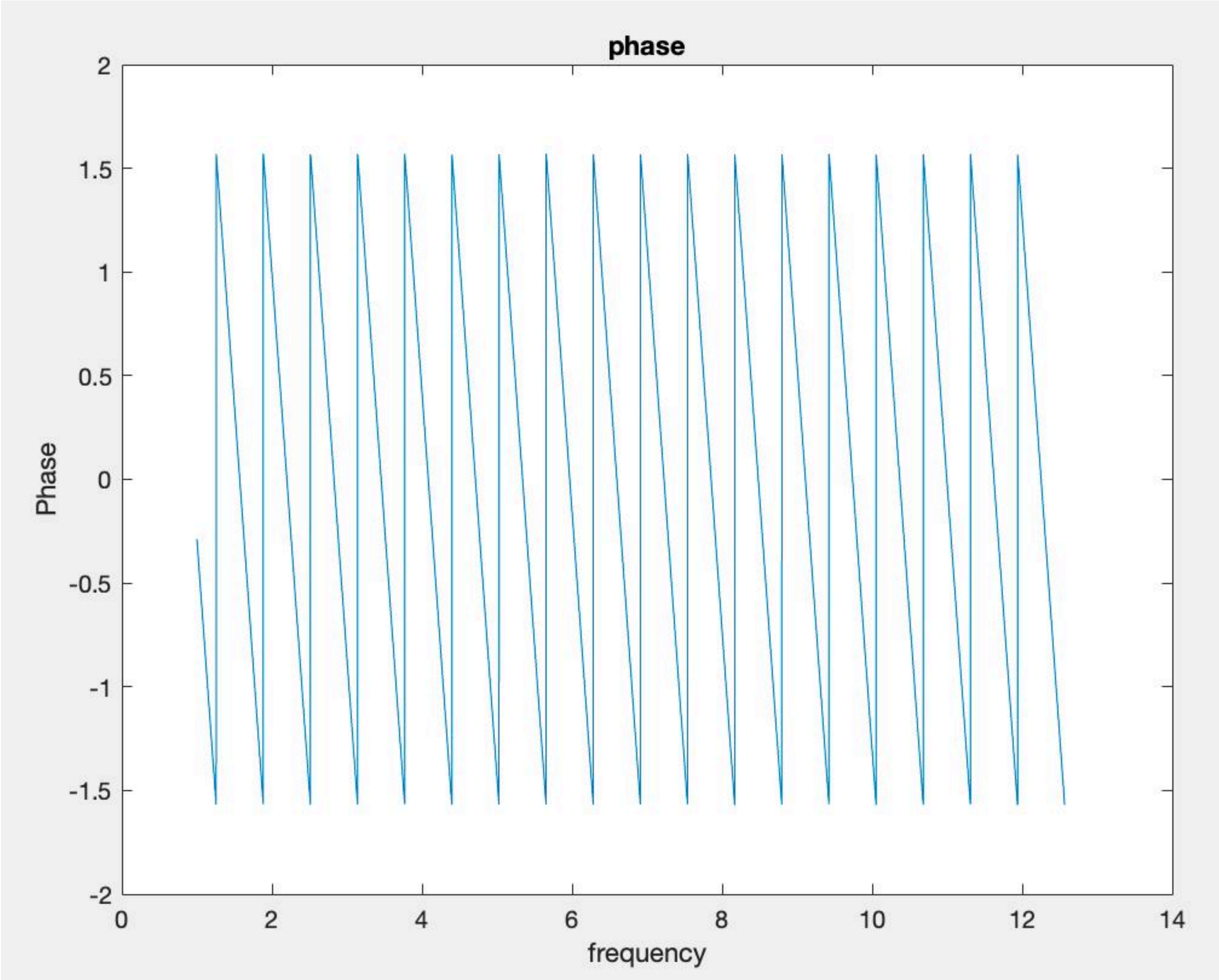
- Submit the labeled plot.

$$
\begin{aligned}
b_0 &= 1 \\
m &= 10 \\
Hcomb_{func} &= @(z) b_0 * (1 - z^{-M}) \\
z &= i * (1 : 10 : 10e^4) \\
for i &= 1 : length(z) \\
H_{comb}(i) &= Hcomb_{func}(z(i)) \\
& end
\end{aligned}
$$

plus plots



Mag
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
Magnitude (dB)
10^{0}
10^{1}
Frequency



b) Remove the three zeros located at $z = e^{j0}, e^{\pm j\frac{\pi}{5}}$. This allows frequencies at $\hat{\omega} = \pm\frac{\pi}{5}$ to pass through without being zeroed out. Plot the magnitude of the normalized frequency response $\left|H_{modified\_comb}(e^{j\hat{\omega}})\right|$ of the modified comb filter. For normalized frequency response, **choose b_0 so that $\left|H_{modified\_comb}(e^{j0})\right|$=1.** You may find conv() in MATLAB a helpful tool to expand the product of polynomials.

- Submit the labeled plot and identify the value of $\left|H_{modified_{comb}}(e^{j\frac{\pi}{5}})\right|$.
- In the final project, the Arduino uses a sampling interval of T_s=10ms (see part 1 of HW 9). What is the corresponding continuous time frequency f in Hz that corresponds to a digital frequency of $\hat{\omega} = \pm\frac{\pi}{5}$?

Remove zeros from $e^{\pm j\frac{\pi}{5}}$ and e^0.

We rewrite but divide by the poles to get

$$\frac{b_0(1 - z^{-M})}{(z - e^{0j})(z - e^{j\frac{\pi}{5}})(z - e^{-j\frac{\pi}{5}})}$$

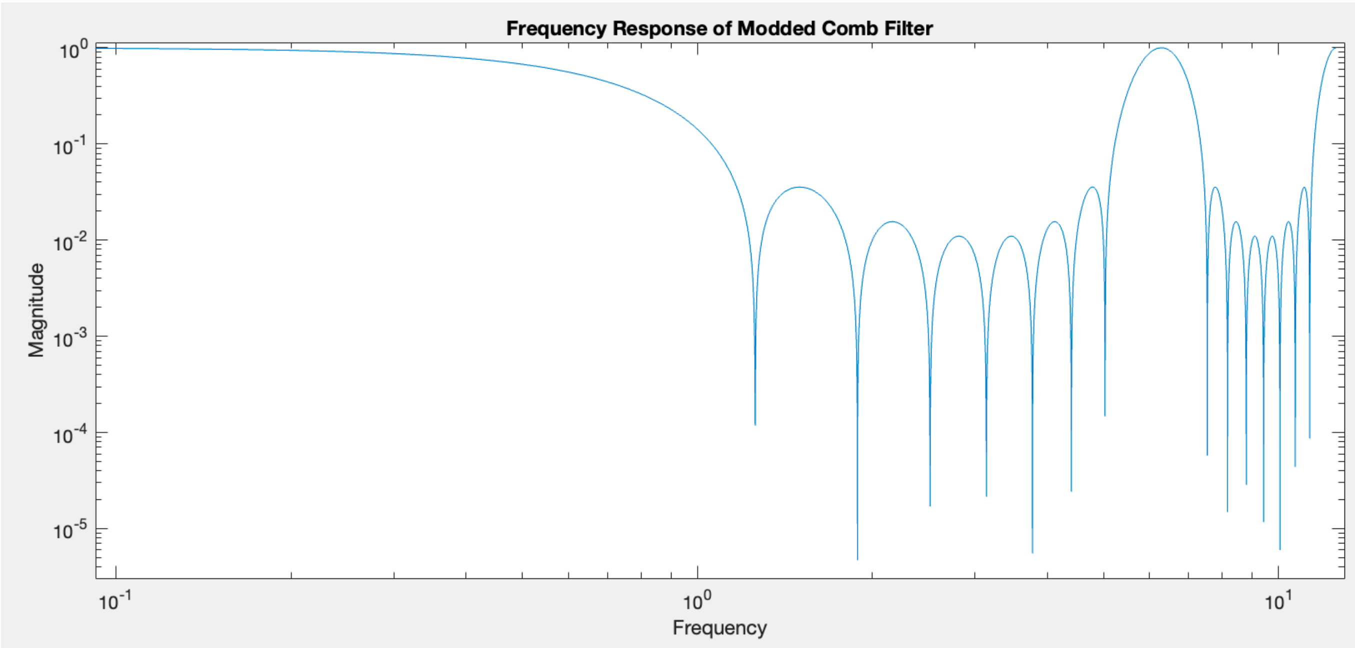
But setting b_0 to be $\frac{1}{26.1803}$ (which is the amplitude at 0 if b_0 were 1).

In this new $H_{modified_{comb}}$ function, $e^{j\frac{\pi}{5}} = 0.5258$

A digital frequency of $\pm\frac{\pi}{5}$ rad/sample, and a sampling frequency of 0.010 seconds/sample make

$$\frac{\frac{\pi}{5}}{(2\pi)(0.010)} = 10$$

So with this filter we are keeping a continuous time frequency of 10Hz (cycles/sample)



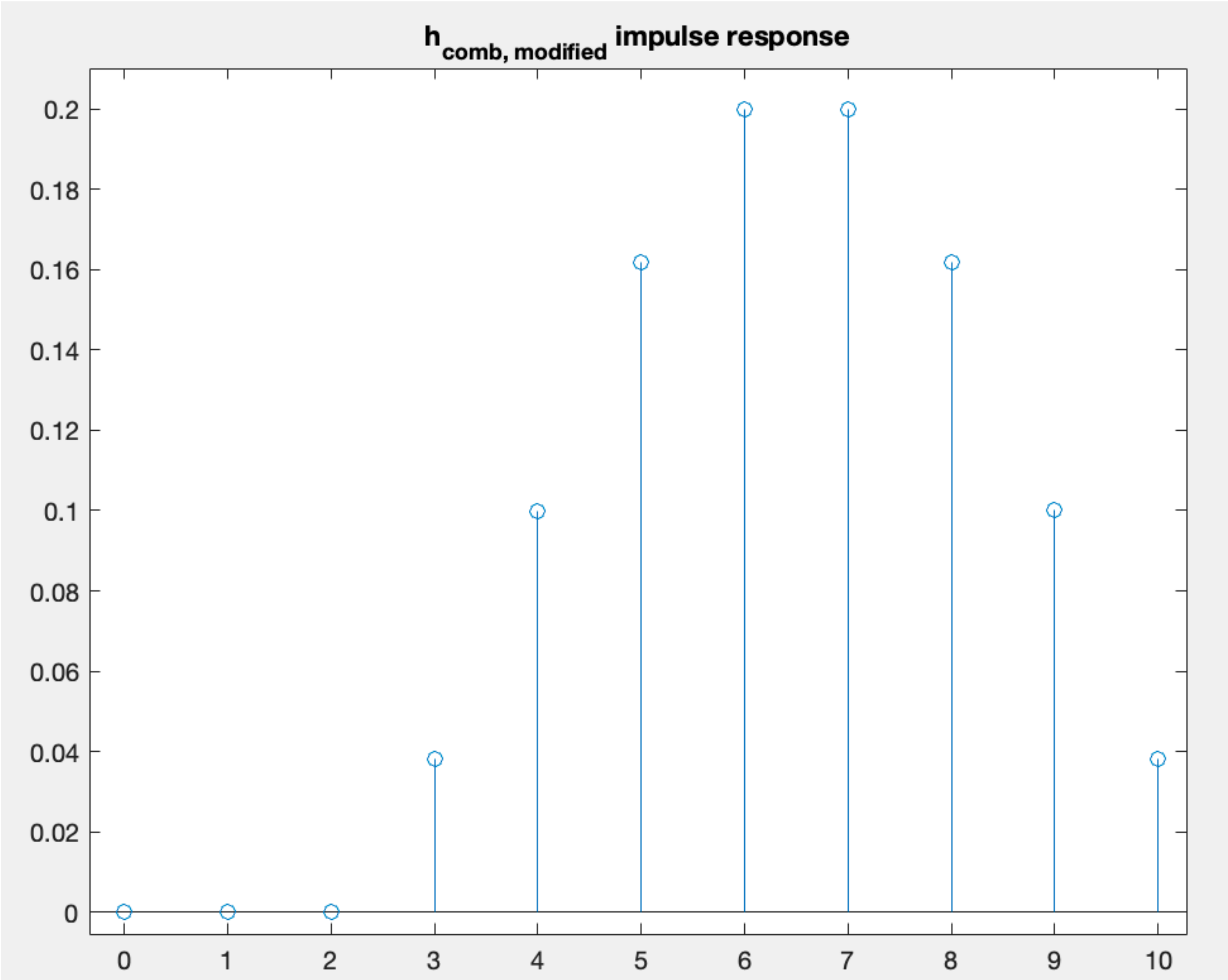
c) Plot the impulse response of the filter $h_{modified\_comb}[n]$. List the corresponding filter coefficients in a table.

- Submit the labeled plot.

We use our modified function and matlab's conv() to see that

$$\frac{b_0(1-z^{-10})}{(z-e^{0j})(z-e^{j\frac{\pi}{5}})(z-e^{-j\frac{\pi}{5}})} = \frac{b_0(1-z^{-10})}{z^3 - 2.6180z^2 + 2.6180z - 1}$$

and run that through a Wolfram Alpha series expansion to obtain the coefficients of the expanded series, which we stick back into a plot



z^0	**2.460E-05**
z^{-1}	9.398E-06
z^{-2}	9.390E-06
z^{-3}	3.820E-02
z^{-4}	1.000E-01
z^{-5}	1.618E-01
z^{-6}	2.000E-01
z^{-7}	2.000E-01
z^{-8}	1.618E-01
z^{-9}	1.000E-01
z^{-10}	3.820E-02